

Lesson 129 (0.5) Ground (1.0) AATD (0.8) Simulated Instrument

Lesson Objective

The purpose of this lesson is to introduce maneuvering solely by reference to flight instruments and introduce basic radio navigation techniques and recoveries from unusual attitudes to facilitate aeronautical decision making and situational awareness.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed:
- Review Maneuvers and Procedures to be Performed in the Simulator
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05: Maintaining Aircraft Control: Upset Prevention and Recovery Training.
- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
 - Chapter 6: Airplane Attitude Instrument Flying Using Analog Instrumentation/Electronic Flight Display
 - Chapter 7: Basic Flight Maneuvers using Analog/Electronic Flight Display
 - Chapter 9: Navigation Systems.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 16: Navigation.
- Aeronautical Information Manual. ([AIM](#))
 - Chapter 1: Air Navigation.

Instructor Guidance

- This is the student's first simulator lesson dedicated entirely to instrument flying. Before putting on the view-limiting device, ensure the student has a solid conceptual understanding of why VFR pilots need these skills — the most likely real-world scenario is inadvertent IMC entry. Frame this lesson around that context from the start.
- Quiz the student during the pre-brief on instrument cross-check technique, fixation and omission errors, and the control and performance method. A student who cannot explain the radial scan or why fixating on a single instrument is dangerous is not conceptually ready for the simulator session — address these gaps before proceeding.
- Emphasize from the very beginning that the student must trust the instruments over their physical senses. Spatial disorientation is a genuine physiological phenomenon — not a failure of attention — and understanding this distinction helps students override the instinct to "feel" their way through instrument flight.
- For straight and level flight under the hood, watch closely for instrument fixation. The most common early error is staring at the altimeter or airspeed indicator while neglecting the

attitude indicator. Reinforce the radial scan — attitude indicator as the hub, supporting instruments as spokes — consistently throughout the session.

- Trim is as critical under the hood as it is in visual flight. If the student is muscling the aircraft during instrument maneuvers rather than trimming to relieve pressure, stop and correct it. Students who develop poor trim habits during instrument training carry those habits into every subsequent flight.
- For unusual attitude recoveries, strongly emphasize the wings-level-before-pitch-up rule for the nose-low recovery. Students who pull back while still in a bank during a nose-low unusual attitude can enter an accelerated stall or exceed structural limits — this is the most dangerous error in unusual attitude recovery and must be understood and demonstrated clearly before the student attempts it.
- For VOR and GPS navigation, take your time with the avionics setup steps. The G430 and Avidyne interface differ between aircraft, and students who are unclear on which button switches between GPS and VLOC will struggle in the air. Work through the steps slowly in the simulator where errors have no consequences.
- When covering the pitot-static blockage table, use real-world scenarios to make each failure mode memorable. For example, ask the student what they would think if their airspeed read zero on takeoff — would they abort? This scenario-based approach is more effective than rote memorization of the table.
- Use the simulator's ability to pause and replay as a teaching tool. If the student develops an unusual attitude during an exercise, pause, talk through what happened and why, then reset and repeat. The simulator environment uniquely supports this iterative teaching approach.

Lesson Content (Pre-Brief)

Review Maneuvers and Procedures to be Performed in the Simulator

- Quiz student on why pilots need to know these techniques even if flying VFR and necessary steps to take if encountering IMC.
 - Emphasize positive control of the aircraft and trusting what the flight instruments are showing.
 - Contact ATC.
 - Exit IMC by descending (if able) or doing a 180 degree turn.
- Ask student how fixation and omission affect flight in IMC conditions
 - Quiz student on how to do an instrument cross-check.
 - Instrument interpretation.
- Discuss the control and performance method for straight and level, climbs, descents, and turns to headings:
 - Pitch + Power = performance.
 - Setting a certain pitch with a power setting results in the desired performance.
 - For example: around 2.5 degrees of pitch + around 2300 RPM = straight and level flight at 90 knots.
 - For example: around 8-10 degrees of pitch + full RPM = climb at 79 knots.
- VOR and GPS navigation procedures:

- Refer to the VOR and GPS sections in the lesson content section below.

Lesson Content (AATD)

Straight-and-Level Flight (Instrument)

- Establish and maintain constant heading and altitude using attitude + power = performance.
 - Around 2.5 degrees of pitch + around 2300 RPM = Straight and Level at 90kts
- The attitude indicator is primary for pitch and primary for bank when the aircraft is stabilized.
- Pitch control:
 - Adjust pitch to maintain constant altitude (small, smooth yoke movements).
 - Lower pitch attitude + higher RPM = higher airspeed
 - Higher pitch attitude + lower RPM = lower airspeed
 - Reference the altimeter and VSI to confirm minor trends.
 - Avoid chasing the VSI — use it as a trend instrument, not a control reference.
- Bank control:
 - Use the heading indicator to maintain heading.
 - Cross-check the turn coordinator or ball to ensure coordination and detect small banks early.
 - Make small aileron and rudder corrections together to maintain wings level.
- Power control:
 - Set and hold the power setting appropriate for the desired airspeed (e.g., 2300–2400 RPM in a Warrior).
 - Use airspeed indicator as the primary power instrument in level flight.
- Trim:
 - Trim off all control pressures once altitude and airspeed stabilize.
 - Proper trim reduces workload and helps maintain precise control.
 - Instrument Cross-Check and Interpretation:
 - Begin with a selective radial scan — attitude indicator as hub; supporting instruments as spokes.
 - Correct for small deviations promptly and smoothly (1/2-bar width corrections on the attitude indicator).
 - Verify performance by comparing multiple instruments (altimeter, VSI, airspeed).
- Common tendencies:
 - Over-controlling due to fixation on one instrument.
 - Failure to recognize small bank or pitch changes early.
 - Slow power response leading to airspeed/altitude drift.
- Emphasize the need to scan!

Constant Airspeed Climbs (Instrument)

- Use attitude + power = performance; pitch controls airspeed, and power controls rate.
- Select a specific pitch attitude on the attitude indicator that yields the desired airspeed
 - V_Y : About a 10 degrees pitch attitude + full power = climb at 79 knots.
 - V_{CLIMB} : About a 5 - 7.5 degrees of pitch attitude + full power = climb at 87 knots.

- Apply smooth power to full while maintaining coordinated flight with rudder.
- Cross-check primarily with the attitude indicator, and reference supporting instruments:
 - Airspeed indicator – confirms constant climb speed.
 - VSI and altimeter – confirm climb performance trend.
 - Heading indicator + turn coordinator – ensure wings level.
- Trim off control pressures once the airspeeds stabilizes.
- Anticipate level-off approximately 10% of the rate of climb (for example, 50 feet per 500 fpm).
- Transition by reducing pitch and power smoothly to cruise settings while maintaining constant airspeed.

Constant Airspeed Descents (Instrument)

- Use attitude + power = performance; pitch controls airspeed, and power controls rate.
- Select a specific pitch attitude on the attitude indicator that yields the desired airspeed, adjusting power and pitch for desired airspeed
 - Nose right under the horizon line (attitude indicator) + power idle = glide descent at 73kts
 - Nose at -2.5 degrees pitch (attitude indicator) + 1600-1800 RPM = descent at 90kts
- Adjust power to desired RPM while maintaining coordinated flight with rudder.
- Cross-check primarily with the attitude indicator, and reference supporting instruments:
 - Airspeed indicator – confirms constant descent.
 - VSI and altimeter – confirm descent performance trend.
 - Heading indicator + turn coordinator – ensure wings level.
- Trim off control pressures once the airspeed stabilizes.
- Anticipate level-off approximately 10% of the rate of climb (e.g., 50 feet per -500 fpm).
- Transition to level flight by increasing pitch and power smoothly to cruise settings while maintaining constant airspeed.

Turns to Headings (Instrument)

- Establish appropriate bank angle for a standard rate turn
 - To calculate how many degrees of bank a standard rate of turn is, use the following formula:
 - Divide airspeed by 10 and add ½ of the result.
 - Warrior standard rate:
 - $90/10 = 9$.
 - $9/2 = 4.5$.
 - $4.5 + 9 = 13.5$ degrees = standard rate.
- Use the attitude indicator to set and hold bank angle; turn coordinator confirms standard-rate turn (3 degrees per second)
 - A 180 degree turn takes 1 minutes and 360 degree turn takes 2 minutes
- Primary instruments:
 - Heading indicator – primary for bank when stabilized.
 - Attitude indicator – establishes bank attitude.
 - Altimeter + VSI – monitor altitude; adjust pitch to maintain level flight.

- Lead the rollout by $\frac{1}{2}$ the bank angle in degrees (e.g., 10° early from a 20° bank).
- Maintain constant power and coordinated rudder throughout turn
 - Remind the student that as the bank increases, pitch must increase to compensate for the reduction in vertical lift.
- Cross-check scan: Attitude → Heading → Altimeter → Turn Coordinator → Repeat.

Recovery from Unusual Flight Attitudes (Instrument)

- Recognize an unusual attitude by instrument interpretation (not “feel”):
 - Nose-high: Increasing pitch, decreasing airspeed, increasing altitude.
 - Nose-low: Decreasing pitch, increasing airspeed, altitude loss, possible turn indication.
- Recovery (Nose-High):
 - Apply maximum power.
 - Lower the nose to prevent stall — relax back pressure smoothly.
 - Level the wings with coordinated aileron and rudder.
 - Return to desired attitude and trim.
- Recovery (Nose-Low):
 - Reduce power to prevent overspeed.
 - Level the wings using coordinated aileron and rudder.
 - Gently raise the nose to the level flight attitude.
 - Re-trim and resume normal scan.
 - EMPHASIZE wings level before pitch up; cover what happens when load factor increases
- Emphasize instrument cross-check (attitude → airspeed → altimeter → VSI → heading → turn coordinator).
- Avoid over-controlling; make small, smooth corrections.
- Reinforce the “power + attitude + trim” relationship throughout recovery.

Radio Communications, Navigation Systems/Facilities, and Radar Services (Instrument)

- Radio Communication
 - Use standard phraseology and proper radio technique (listen before transmitting, think before keying).
 - Ensure clear, concise, and professional transmissions — include who you’re calling, who you are, where you are, and what you want.
 - Acknowledge and read back critical instructions (altitude, heading, altimeter settings, runway assignments).
 - Maintain situational awareness by monitoring guard (121.5) or ATIS/CTAF as appropriate.
 - Always verify frequency changes; anticipate upcoming ATC handoffs or frequency zones.
 - Discuss radio calls between the pilot and ATC when inadvertently entering IMC conditions.
- Navigation Systems/Facilities
 - Discuss how GPS can be used after encountering IMC conditions
 - Using nearest page, NAV, and AUX page
 - Demonstrate various ways of proceeding direct to an airport
 - Direct button → input the airport identifier → enter → enter.

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- Nearest page → nearest airports → direct → enter → enter.
- For the G5:
 - Press the CDI button on the G430 to switch it from VLOC to GPS.
 - The G5 HSI should then show the magenta GPS needle.
- For the Avidyne:
 - Press the up-most left bezel key on the PFD. This will cycle between GPS1, VLOC1, GPS2, and VLOC2. Select the desired GPS.
- For the purpose of this lesson, the student should be able to:
 - Proceed direct to an airport.
 - Intercept and track a GPS course.
- Discuss how instrument approaches work (surface level)
- Demonstrate how to use the VOR.
 - Set the VOR frequency in the G430 or G650 (show where to find the frequency).
 - Tune and ID the VOR by pressing the ID button and NAV1 or NAV2 (depending on which NAV is being used).
 - Verify with the morse code on a sectional chart.
 - For the G5:
 - Once tune and ID is complete, press the CDI button on the G430 or G650 to switch it from GPS to VLOC.
 - The G5 HSI should then show the green VOR needle.
 - To adjust the course, press the right knob on G5, scroll to the right to CRS, and press the right knob. This allows the pilot to twist to the desired course.
 - For the Avidyne:
 - Once tune and ID is complete, press the up-most left bezel key on the PFD. This will cycle between GPS1, VLOC1, GPS2, and VLOC2. Select the desired VLOC.
 - To adjust the course, use the left knob on the PFD to adjust the needle to the desired course. To center it, simply press the left knob.
 - Cover the difference between an outbound and inbound indication.
 - Outbound: arrows point away from each other.
 - Inbound: arrows point the same way.
 - When given an outbound instruction like "TCF609 intercept and track the 360-radial outbound", the course should be set to 360.
 - When given an inbound instruction like "TCF609 intercept and track the 360-radial inbound", the tail should be set to 360 OR take the reciprocal of 360 and set that number to the course.
 - For the purpose of this lesson, the student should be able to:
 - Figure out the current radial.
 - Intercept and track inbound/outbound instructions using an intercept angle of 30-45 degrees in the correct direction.
- Radar Services
 - Understand primary radar (range and azimuth only) and secondary radar (transponder-based — Mode A/C/S).

- Transponder codes:
 - 1200 – VFR, 7500 – Hijack, 7600 – Radio Failure, 7700 – Emergency.
- When working with ATC radar:
 - Listen for assigned discrete transponder code.
 - Respond to radar vectors using standard-rate turns and precise altitude control.
 - Read back headings and altitudes to ensure clarity.

Instrument errors

- Cover how to recognize an instrument error/failure and how to solve it
 - Pitot-static blockages:

Situation / Instrument	Airspeed	Altimeter	VSI
Blocked pitot	Zero	Works	Works
Blocked pitot and drain hole	High in climb Low in descent	Works	Works
Blocked static	Low in climb High in descent	Frozen	Frozen
Alternate static	Reads high	Reads high	Momentary climb

- ADAHRS failure:
 - Red X's.
 - Reset the circuit breaker (has to be done within 30 seconds of the AHRS failure).
 - If PFD function does not return then use the standby instruments, contact ATC, and land as soon as practical.
- DAU failure (Avidyne):
 - Red X's.
 - Reset the circuit breaker.
 - If MFD function does not return then contact ATC, and land as soon as practical.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 130.
- Assign homework:

Completion Standards

This lesson is complete when the applicant can, with instructor guidance:

1. Use radio aids for orientation.
2. Track and bracket radials.

3. Identify and practice proper corrective actions to unusual attitudes.
4. Maintain altitude ± 200 ft, heading ± 30 degrees, and airspeed ± 10 knots.